SCHEDULING, AGGREGATE PLANNING & MPS

Exam-Focused Complete Revision Notes

Based on the uploaded 25-page course notes. Includes theory, comparisons, formulas, numerical methods, examples, and last-minute revision points.

1. SCHEDULING — CORE IDEA

Definition: Scheduling is deciding when and in what sequence production activities, jobs, operations, or projects should be performed efficiently.

It answers: What? How much? When? On which machine/resource?

Objectives: meet delivery dates; maximize resource utilization; minimize production time and idle time; minimize WIP; reduce production cost; ensure smooth flow; maintain flexibility.

Flexibility is needed for: demand changes, machine breakdowns, material shortages, absenteeism, and urgent orders.

2. PRODUCTION PLANNING HIERARCHY

Demand Forecast → Aggregate Plan → MPS → MRP → Detailed Scheduling → Dispatching → Production

Level	Main activity	Typical horizon
Strategic	Capacity/facility planning	Several years
Tactical	Aggregate planning	6–18 months
Intermediate	Master Production Schedule	Weeks/months
Operational	Detailed scheduling	Days/weeks
Shop floor	Dispatching & sequencing	Hours/days

3. AGGREGATE PLANNING

Definition: Medium-term plan determining overall production quantity, workforce, inventory, and capacity utilization. Products are considered in groups/families rather than individually.

Objectives: satisfy demand; use capacity effectively; determine workforce; maintain reasonable inventory; minimize total cost; determine overtime/subcontracting; provide a basis for MPS; balance demand and capacity.

Inputs: demand forecast, beginning inventory, production capacity, workforce information, production costs, and company policies.

Important costs: regular labour, overtime, hiring, layoff, inventory holding, subcontracting, shortage/backorder.

4. AGGREGATE PLANNING STRATEGIES

Hiring & Layoff: hire when demand rises; lay off when demand falls. Workforce follows demand, but hiring/layoff costs and dissatisfaction can occur.

Overtime & Undertime: overtime during high demand and fewer hours during low demand. Avoids frequent hiring but overtime is expensive.

Subcontracting: outsource part of production to another company.

Additional shifts: add a second or third shift when demand increases.

Inventory strategy: produce extra during low demand and store it for future demand. Disadvantages include storage, insurance, obsolescence, and tied-up capital.

5. LEVEL vs CHASE STRATEGY

Feature	LEVEL	CHASE
Production	Relatively constant	Changes with demand
Workforce	Stable	Changes
Inventory	Usually higher	Usually lower
Hiring/Layoff	Low	High
Employee stability	High	Low
Flexibility	Moderate	High
Best suited when	Stable workforce important	Demand varies significantly

Memory: LEVEL = keep production level/same. CHASE = production chases/follows demand.

6. MASTER PRODUCTION SCHEDULE (MPS)

Definition: A detailed production plan specifying which finished products are produced, in what quantity, and in which time period.

MPS answers: What finished product? How many? When?

Purpose: determine finished-product quantities and timing; provide input to MRP; coordinate customer orders and production; use capacity effectively; provide delivery information; coordinate manufacturing, purchasing, and inventory.

Inputs: aggregate production plan, customer orders, demand forecast, beginning inventory, capacity information, and current production status.

Outputs: production quantities, dates, finished-product requirements, available-to-promise information, MRP input, and capacity requirements.

Key distinction: MPS = finished products. MRP = components/materials.

7. MPS vs MRP — EXAM ANSWER

MPS	MRP
Finished products	Components/materials
What/how many/when to make	What components/how many/when required
Example: 500 bicycles in Week 4	500 frames, 500 seats, 1000 wheels, etc.

8. JOB SHOP PRODUCTION & SCHEDULING

Job shop: low-volume production where jobs may require different sequences of operations.

Examples: tool rooms, repair workshops, custom machinery, die/mould manufacturing, prototypes.

Characteristics: low volume, high variety, customized products, different routes, general-purpose machines, skilled workers, complex scheduling, potentially high WIP, longer lead time, variable machine utilization.

Scheduling decisions: which job, which machine, sequence, start time, completion time.

9. DISPATCHING / SCHEDULING RULES

FCFS — First Come First Served: first arrival is processed first. Simple, but may give poor due-date performance.

SPT — Shortest Processing Time: shortest job first. Reduces average flow time and WIP, but long jobs can be delayed.

EDD — Earliest Due Date: earliest due date first. Useful for delivery commitments.

LPT — Longest Processing Time: longest processing time first.

Critical Ratio:

$$CR = (\text{Time remaining until due date}) / (\text{Processing time remaining})$$

CR < 1: behind schedule. **CR = 1:** exactly on schedule. **CR > 1:** schedule margin.

10. DISPATCHING

Dispatching is authorizing/releasing a job for production according to the established schedule. The dispatcher decides the job, machine, operator, and start time.

11. MASS PRODUCTION

Definition: large quantities of standardized products using specialized equipment and repetitive processes.

Examples: automobiles, appliances, electronic products, bottled products, standardized components.

Characteristics: high volume, low variety, standardized products, specialized machines, product layout, repetitive operations, low unit cost, high mechanization/automation, short cycle, continuous material flow.

Scheduling focuses on: production rate, line balancing, machine/material availability, cycle time, workforce allocation, maintenance, and delivery.

12. LINE BALANCING — FORMULAS & NUMERICAL

Definition: assigning tasks to workstations so workload is as evenly distributed as possible.

Objective: minimize idle time, bottlenecks, and production delays while meeting the required production rate.

Cycle Time:

$$C = \text{Available Production Time} / \text{Required Production Quantity}$$

Theoretical Minimum Workstations:

$$N_{min} = \Sigma t_i / C$$

Always round N_{min} UP to the next whole number.

Example: 480 cars in an 8-hour shift. Available time = 8 × 60 = 480 min. Cycle time = 480/480 = **1 min/car**.

Exam method: 1) Convert available time to the same unit as demand. 2) Calculate cycle time. 3) Add all task times. 4) Divide total task time by cycle time. 5) Round up.

13. CONTINUOUS PRODUCTION

Definition: materials flow continuously with very little interruption.

Examples: oil refineries, cement, chemical, steel, paper, power generation, petrochemical industries.

Characteristics: very high volume, very low variety, continuous flow, high automation, specialized equipment, high capital investment, long operating periods, costly interruptions, preventive maintenance, detailed process control.

Scheduling focus: production rate, raw-material availability, equipment availability, maintenance, product changeover, quality control.

14. BATCH PRODUCTION

Batch production lies between job shop and mass/continuous production. Example: pharmaceutical batches of different products. Scheduling must determine the sequence of batches and changeover times.

15. PROJECT PRODUCTION

Definition: producing a unique product or completing a unique activity at a specific location within a defined time and budget.

Examples: bridge, building, ship, aircraft development, power plant installation, new manufacturing facility.

Characteristics: unique output, large scale, long duration, high investment, resources move to the product/location, interdependent activities, extensive planning, critical time/cost control, high risk, project management techniques.

Layout: fixed-position — the product remains in one place and resources are brought to it.

16. PROJECT SCHEDULING TECHNIQUES

Gantt Chart: activities against time; shows start, finish, duration, progress, and overlap.

PERT: Program Evaluation and Review Technique; useful when activity times are uncertain.

CPM: Critical Path Method; used when activity times are relatively known and the critical path must be identified.

Network analysis: shows relationships among project activities.

17. PERT — VERY IMPORTANT NUMERICAL

Three estimates: Optimistic (a) = minimum; Most likely (m) = realistic; Pessimistic (b) = maximum.

Expected time:

$$t_e = (a + 4m + b) / 6$$

Example: a = 2 days, m = 5 days, b = 8 days.

t_e = [2 + 4(5) + 8]/6 = 30/6 = **5 days**.

Exam steps: write a, m, b → substitute → multiply m by 4 → add → divide by 6 → write units.

18. CPM & CRITICAL PATH — VERY IMPORTANT

Critical Path: the longest path through the project network. It determines the minimum possible completion time.

Critical activities have: Float = 0.

Important: delaying a critical activity can delay the project unless corrective action is taken.

Example: A→B→D = 4+5+3 = 12 days. A→C→D = 4+2+3 = 9 days. Therefore Critical Path = **A→B→D**; project duration = **12 days**.

CPM exam rule: Draw network → identify every complete path → add activity durations on each path → choose the LONGEST path → that is the critical path → its duration is project duration.

19. PRODUCTION TYPE COMPARISON

Feature	Job Shop	Mass	Continuous	Project
Volume	Low	High	Very high	One/very low
Variety	High	Low	Very low	Unique
Product	Customized	Standardized	Standardized	Unique
Equipment	General purpose	Specialized	Highly specialized	Project-specific
Layout	Process	Product	Product/process flow	Fixed-position
Scheduling	Job sequencing	Line balancing	Flow/control	PERT/CPM

20. SCHEDULING PERFORMANCE MEASURES

Makespan: total time to complete all jobs.

Flow Time: Completion Time – Arrival Time.

Tardiness: T = max(0, C – d), where C = completion time and d = due date.

Machine Utilization: Actual Operating Time / Available Time × 100.

Average Waiting Time: average time jobs wait before processing.

On-time Delivery: percentage of orders completed by promised date.

21. PLANNING vs SCHEDULING

Planning	Scheduling
What should be done	When and in what sequence
Broader activity	More detailed activity
Longer-term	Shorter-term
Determines requirements	Allocates resources and time
Example: 10,000 units/month	Example: 500 units on Monday

22. LAST-MINUTE FORMULA SHEET

Critical Ratio

$$CR = \text{Time remaining until due date} / \text{Processing time remaining}$$

Cycle Time

$$C = \text{Available Production Time} / \text{Required Production Quantity}$$

Minimum Workstations

$$N_{min} = \Sigma t_i / C \rightarrow \text{round UP}$$

PERT Expected Time

$$t_e = (a + 4m + b) / 6$$

Flow Time

$$\text{Flow Time} = \text{Completion Time} - \text{Arrival Time}$$

Tardiness

$$T = \max(0, C - d)$$

Machine Utilization

$$\text{Utilization} = \text{Actual Operating Time} / \text{Available Time} \times 100$$

23. ONE-PAGE MEMORY MAP

AGGREGATE PLAN → Overall production level

MPS → Specific finished products + quantities + timing

MRP → Components + materials

JOB SHOP → Low volume + high variety → sequencing

MASS → High volume + low variety → line balancing

CONTINUOUS → Very high volume + very low variety → flow/process control

PROJECT → Unique + long duration → Gantt/PERT/CPM

LEVEL → constant production/workforce → inventory absorbs demand changes

CHASE → production follows demand → lower inventory, more workforce changes

PERT → uncertain times → 3 estimates

CPM → known/relatively known times → longest path = critical path

Critical path → longest path + zero float

24. EXAM WRITING TEMPLATES

For a 2-mark definition: Write the definition + one key purpose/example.

For a 5-mark theory question: Definition → 4–6 points → example → short conclusion.

For Level vs Chase: draw the comparison table; it is faster and clearer.

For PERT: write formula first, substitute values, calculate, include units.

For CPM: draw the network neatly, list all paths, calculate each path duration, circle the longest path, state project duration.

For line balancing: convert time units → calculate cycle time → calculate total task time → calculate Nmin → round upward.

25. TOP PRIORITY FOR TOMORROW'S EXAM ■■■

1. CPM: drawing the network, finding all paths, critical path, project duration, float.
2. PERT: a, m, b and expected time calculation.
3. Line balancing: cycle time and minimum workstations.
4. Critical Ratio and interpretation.
5. FCFS, SPT, EDD, LPT definitions and advantages/disadvantages.
6. Aggregate planning strategies.
7. Level vs Chase strategy.
8. MPS vs MRP.
9. Job shop vs mass vs continuous vs project production.
10. Planning vs scheduling and the complete production-planning hierarchy.

26. 30-SECOND FINAL REVISION

Aggregate → MPS → MRP → Scheduling → Dispatching → Production

Level = Same | Chase = Follow | MPS = Product | MRP = Parts | Job = Sequence | Mass = Balance | Continuous = Flow | Project = PERT/CPM

PERT: (a+4m+b)/6 **CPM:** Longest path **Critical:** Float = 0